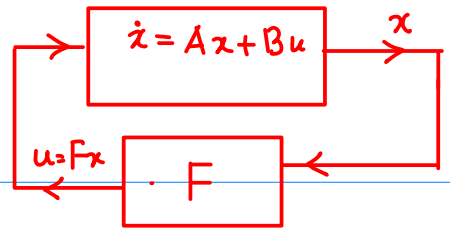
L7

19/09



$$x(t) = e^{\boxed{At}} x_0 + \int_0^t e^{A(t-\tau)} B u(\tau)$$

$$u(t) = F x(t)$$

$$\dot{x} = Ax + Bu$$

$$\dot{x}(t) = (A + BF)\, x(t) \Rightarrow x(t) = e^{(A+BF)t} \underline{\underline{x_0}}$$

e.g.

$$A = \begin{bmatrix} 5 & 2 & 1 \\ -1 & 4 & 1 \\ 1 & 0 & 3 \end{bmatrix} \qquad b = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$\sigma(A) = \{4, 4, 4\}$ Unstable $\mathbb{C}^+$

Q: (I) Can the system be made stable?

(II) Can we place the eigenvalues of the system at $\{-1, -2, -3\}$

$$\rank(\mathcal{C}_{AB}) = 3 \Rightarrow \text{Controllable}$$

C.C.F: $T = PW^{-1} = \mathcal{C}_{Ab}\, \mathcal{C}_{\tilde{A}\tilde{b}}^{-1}$

$$\begin{tikzcd} \mathcal{B}_1 = \{e_1, e_2, e_3\} \; \mathbb{R}^3 \arrow[r, "A+bF"] & \mathbb{R}^3 \\ \mathbb{R}^3 \arrow[u, "T"] \arrow[r, "\tilde{A}+\tilde{b}\tilde{F}"] & \mathbb{R}^3 \arrow[u, "T"'] \end{tikzcd}$$

$\mathcal{B}_2 = \{\ \}$

$$\chi_A(s) = (s-4)^3 = s^3 - 12s^2 + 48s - 64$$
$$= \chi_{\tilde{A}}(s)$$

$$\tilde{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 64 & -48 & 12 \end{bmatrix} \qquad \tilde{b} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$T = \begin{bmatrix} -20 & 10 & -1 \\ 4 & -6 & 1 \\ 28 & -10 & 1 \end{bmatrix}$$

$$\boxed{\underline{\underline{\tilde{A}}} = T^{-1} \underline{\underline{A}}\, T \qquad \tilde{b} = T^{-1} b}$$

$$\tilde{F} = [f_1 \quad f_2 \quad f_3]$$

$$\tilde{A} + \tilde{b}\tilde{F} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 64+f_1 & -48+f_2 & 12+f_3 \end{bmatrix}$$

$$\chi_{\tilde{A}+\tilde{b}\tilde{F}} = s^3 + (\underline{-12-f_3})s^2 + (\underline{48-f_2})s + (\underline{-64-f_1})$$
$$= (s+1)(s+2)(s+3) = s^3 + 6s^2 + 11s + 6$$

$$f_1 = -70 \qquad f_2 = 37 \qquad f_3 = -18$$

$$F = \tilde{F} T^{-1} = [-53 \quad -36 \quad -35]$$

$$* \; \dot{x} = Ax + bu \qquad x(t) = T\hat{x}(t)$$

$$T\dot{\hat{x}}(t) = AT\hat{x}(t) + bu(t)$$

$$\Rightarrow \dot{\hat{x}}(t) = T^{-1}AT\hat{x}(t) + \underbrace{T^{-1}b}\, u(t)$$

$$\dot{\hat{x}}(t) = \underline{\underline{\tilde{A}}}\, \hat{x}(t) + \tilde{b}\, u(t)$$

$$u(t) = \tilde{F}\hat{x}(t)$$

$$\therefore \dot{\hat{x}}(t) = (\tilde{A} + \tilde{b}\tilde{F})\hat{x}(t)$$

$$\Rightarrow T^{-1}\dot{x}(t) = (\tilde{A} + \tilde{b}\tilde{F})T^{-1}x(t)$$

$$\Rightarrow \dot{x}(t) = \overset{A}{\boxed{T\tilde{A}T^{-1}}}\, x(t) + \overset{b}{\boxed{T\tilde{b}}}\, \overset{F}{\boxed{\tilde{F}T^{-1}}}\, x(t) \qquad u(t) = Fx(t)$$

$$u = Fx = \begin{bmatrix} -53 & -36 & -35 \end{bmatrix} x$$

$$\dot{x} = (A + bF)x \qquad \sigma(A+BF) = \{-1, -2, -3\}$$

$$\Rightarrow x(t) = e^{(A+bF)t} x_0$$

$$= e^{PDP^{-1}t} x_0$$

$$\Rightarrow x(t) = P e^{Dt} P^{-1} x_0$$

$$= P \begin{bmatrix} e^{-t} & & \\ & e^{-2t} & \\ & & e^{-3t} \end{bmatrix} P^{-1} x_0$$

$$= \begin{bmatrix} * \\ * \\ * \end{bmatrix}_{3\times 1}$$

$\exists P$ s.t

$$P^{-1}(A+bF)P = \underbrace{\begin{bmatrix} -1 & & \\ & -2 & \\ & & -3 \end{bmatrix}}_{D}$$

$$\Rightarrow (A+bF) = PDP^{-1}$$

$$\dot{x} = Ax + Bu$$

$\boxed{}$ Suppose $b \in \mathbb{R}^{n\times 1}$. The eigenvalues $(A+bF)$ can be arbitrarily assigned if and only if (A,b) is controllable.

<u>Proof</u>: <u>Only if</u> $\sigma(A+bF)$ can be arb. assigned

To the contr, (A, b) is uncontrollable $\qquad \sigma(A) = \sigma(\tilde{A}_{11}) \cup \sigma(\tilde{A}_{22})$

$$K := \left(\begin{bmatrix} \tilde{A}_{11} & \tilde{A}_{12} \\ 0 & \boxed{\tilde{A}_{22}} \end{bmatrix} + \begin{bmatrix} \tilde{b}_1 \\ 0 \end{bmatrix} \begin{bmatrix} F_1 & F_2 \end{bmatrix} \right) = \begin{bmatrix} \boxed{\tilde{A}_{11} + \tilde{b}_1 F_1} & \tilde{A}_{12} + \tilde{b}_1 F_2 \\ 0 & \boxed{\tilde{A}_{22}} \end{bmatrix}$$

$$\sigma(K) = \sigma(\tilde{A}_{11} + \tilde{b}_1 F_1) \cup \sigma(\tilde{A}_{22})$$

You cannot change with F

If $\left(\sigma(A) \cap \mathbb{C}^+\right) \cap \left(\sigma(\tilde{A}_{22})\right) = \phi \Rightarrow$ Stabilizable

Pole placement (MIMO): $B \in \mathbb{R}^{n\times m}$, $A \in \mathbb{R}^{n\times n}$

Wonham
Linear Multivariable Control

Lemma: Suppose (A,B) is c.a.r. If $B_1 \in \mathbb{R}^{n\times 1}$ is any non-zero column of B, then $\exists$ an $\underline{F_1}$ such that $(\underbrace{A+BF_1, B_1})$ is controllable.

Theorem 2.19: The eigenvalues of $A+B\underline{F}$ are freely assignable, by chosing $F \in \mathbb{R}^{m\times n}$ $\iff$ (A,B) is controllable

Proof: (if) Chose a non-zero column of B say b_1

$\exists F_1$ s.t. $(\underline{A+BF_1}, \underline{b_1})$ is controllable

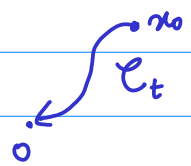
$$\Updownarrow$$

$\exists F_2$ s.t. $\sigma(A+BF_1+b_1\underline{F_2})$ is freely assignable

Finally, $\underline{A}+\underbrace{BF}_{\uparrow} = \underline{A}+\boxed{BF_1+b_1F_2} \in \operatorname{img} B$

Reachability

Controllability



Discrete-time systems

$\mathcal{R}_t \subset \mathcal{C}_t$

Reachable $\overset{\mathcal{C}_{AB}}{\Rightarrow}$ Controll

Controllab $\not\Rightarrow$ Reach

$\boxed{A}$ $\mathcal{C}_{AB}$